Chapter 9

Nonvariational Techniques

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 9; printed p. 495; PDF page 512.

Chinese reference: 中译本对应章：非变分技术，书内第 350 页起。

This editorial study chapter reconstructs Evans's arguments in English, with Chinese mathematical annotations and additional proof steps. Source theorem numbers are separate from the continuous numbering of these notes. Monotonicity, compactness, order, scaling, and convex dissipation provide different mechanisms. Unless specified otherwise, $U \subset \mathbb{R}^n$ is bounded with smooth boundary, $H_0^1(U)$ has norm $\|Du\|_2$, and C_P is its Poincaré constant. This study draft has not undergone independent mathematical review.

9.1 Monotonicity Methods

9.1.1 Structure without a potential

Consider $-\operatorname{div} a(Du) = f$, $u \in H_0^1(U)$, where $f \in L^2(U)$ and $a : \mathbb{R}^n \to \mathbb{R}^n$ is continuous. The source assumes smoothness; existence needs only continuity and

$$\begin{aligned} (a(p) - a(q)) \cdot (p - q) &\geq 0, \\ |a(p)| &\leq C(1 + |p|), \\ a(p) \cdot p &\geq \alpha |p|^2 - \beta, \end{aligned} \tag{9.1}$$

with $\alpha > 0$, $\beta \geq 0$. A weak solution satisfies $\int_U a(Du) \cdot Dv = \int_U fv$ for every $v \in H_0^1(U)$. For $a = DF$ with F convex, monotonicity follows by adding its two supporting-plane inequalities. The converse fails: $a(p) = (I + K)p$ with $K^T = -K \neq 0$ is strongly monotone but has no scalar potential, since its Jacobian is not symmetric. For constant K its skew part cancels from the scalar second-order operator; this example tests the vector-field structure.

9.1.2 Finite-dimensional solvability and energy bounds

Lemma 9.1 (Zeros of a vector field; source §9.1 lemma)
If $V : \overline{B}_r \to \mathbb{R}^m$ is continuous and $V(d) \cdot d \geq 0$ on $|d| = r > 0$, it has a zero in the closed ball.

Expanded proof (the closed ball matters).

If V never vanishes, $T(d) = -rV(d)/|V(d)|$ is a continuous self-map of the closed ball. Brouwer gives $d = T(d)$, hence $|d| = r$ and $r^2 = -rV(d)\cdot d/|V(d)| \leq 0$, a contradiction. Brouwer is the finite-dimensional input from §8.1; an interior zero is not asserted. □

Choose an orthonormal basis (w_j) of $H^1_0(U)$, with dense finite spans X_m. Seek $u_m = \sum_{j=1}^m d_j w_j$ with

$$\int_U a(Du_m) \cdot Dv = \int_U fv \qquad (v \in X_m). \tag{9.2}$$

Theorem 9.2 (Approximation and bounds; source Theorems 1–2)
For every m there is such a u_m, and $\|Du_m\|_2 \leq C_0(1 + \|f\|_2)$, where C_0 depends only on $\alpha, \beta, |U|, C_P$, not on m.

Expanded proof (one coercive estimate does both jobs).

Set $V_j(d) = \int a(\sum d_i Dw_i) \cdot Dw_j - \int f w_j$. Continuity follows by dominated convergence on bounded coefficient sets: the growth bound dominates the integrand by an L^1 function built from the finitely many Dw_i. Orthonormality and Young's inequality give

$$V(d) \cdot d \geq \alpha|d|^2 - \beta|U| - C_P\|f\|_2|d| \geq \tfrac{\alpha}{2}|d|^2 - \beta|U| - \frac{C_P^2}{2\alpha}\|f\|_2^2.$$

A sufficiently large sphere satisfies the lemma, giving a zero and hence (9.2). Test with u_m and repeat the estimate: $\frac{\alpha}{2}\|Du_m\|_2^2 \leq \beta|U| + C_P^2\|f\|_2^2/(2\alpha)$. This also bounds $a(Du_m)$ in $L^2(U;\mathbb{R}^n)$. □

单调不等于梯度

对 $a(p) = (I + K)p$，单调性二次型为 $|p - q|^2$。反对称部分不影响它，却阻止 Da 成为 Hessian。

强制性与单调性分工

强制性先给有限维零点和一致范数界；单调性随后识别极限。弱紧性本身不能推出非线性通量的收敛。

9.1.3 Minty's limit argument

Theorem 9.3 (Weak existence; source Theorem 3)
Under (9.1)*, the Dirichlet problem has a weak solution.*

Expanded proof (eliminate the product of two weak limits).

Extract $u_m \rightharpoonup u$ in H_0^1 and $a(Du_m) \rightharpoonup \xi$ in L^2. For each fixed w_j, the Galerkin identity holds for all sufficiently large m. Passing to the limit and then using density yields $\int \xi \cdot Dv = \int fv$ for all $v \in H_0^1$. Moreover $\int a(Du_m) \cdot Du_m = \int fu_m \to \int fu = \int \xi \cdot Du$. This is an energy identity, not a general rule for products of weak limits.

For fixed $w \in H_0^1$, integrate monotonicity and expand the product. Use the energy identity for its only problematic term to obtain

$$\int_U (\xi - a(Dw)) \cdot (Du - Dw) \geq 0.$$

Set $w = u - tv$ with $t > 0$ and divide by t. As $t \downarrow 0$, $a(Du - tDv) \to a(Du)$ strongly in L^2: continuity gives pointwise convergence, and for $t \leq 1$ its square is dominated by $C(1 + |Du|^2 + |Dv|^2)$. Thus $\int (\xi - a(Du)) \cdot Dv \geq 0$. Replace v by $-v$ for equality. Together with the identity for ξ this proves the weak equation. Only equality on gradients is required; testing solely with gradients has not shown pointwise $\xi = a(Du)$. □

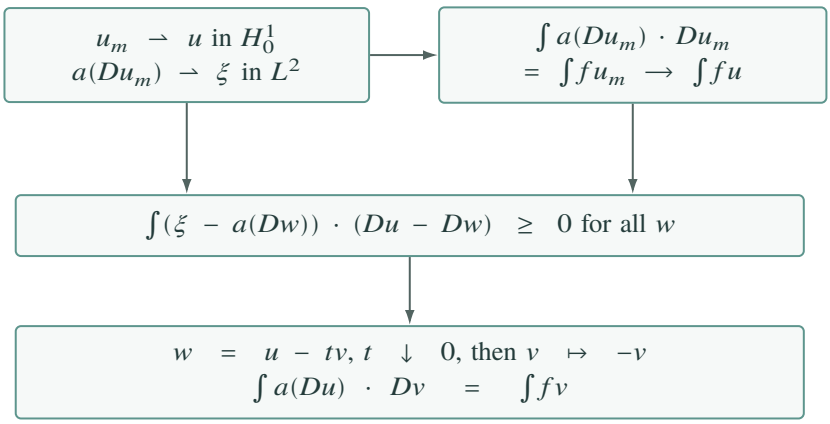


Diagram T9.1. Energy closes the weak-limit argument. The flux limit is initially named ξ. The Galerkin energy identity removes the weak–weak product; monotonicity with arbitrary w, followed by $w = u - tv$, identifies the weak equation. Each arrow requires its own justified limit.

9.1.4 Uniqueness and the regularity hypothesis

Theorem 9.4 (Strong monotonicity; source Theorem 4 and regularity remark)
If $(a(p)-a(q))\cdot(p-q)\ge\theta|p-q|^2$ with $\theta>0$, the weak solution is unique. If also $a\in C^1$, $|Da|\le M$, and ∂U is C^2, then $u\in H^2(U)$.

Expanded proof (uniqueness and difference quotients).

Subtract the equations for $u,\tilde u$ and test by $u-\tilde u$. Then $\theta\|D(u-\tilde u)\|_2^2\le 0$, and Poincaré gives equality. For regularity, difference quotients of $a(Du)$ have the form

$$D_h a(Du)=B_h D_h Du,\qquad B_h=\int_0^1 Da(Du+s(Du(\cdot+he)-Du))\,ds.$$

Here $B_h\eta\cdot\eta\ge\theta|\eta|^2$ and $|B_h|\le M$. Testing by $-D_{-h}(\zeta^2 D_h u)$ gives, after Cauchy–Young,

$$\int\zeta^2|D_hDu|^2\le C(\|f\|_2^2+\|Du\|_2^2).$$

The forcing term is bounded by $\|f\|_2\|D(\zeta^2D_hu)\|_2$; no derivative of f is taken. The difference-quotient criterion gives interior second derivatives. In a C^2 boundary chart the tangential estimate is the boundary argument of §6.3 (or §8.3 with nonsymmetric B_h). The transformed equation recovers the normal second derivative by division by its coefficient, bounded below by ellipticity. Bounded chart derivatives produce lower-order terms controlled by $\|Du\|_2$. A finite covering proves global H^2. This last step explicitly uses the boundary difference-quotient result developed in those chapters. □

只需识别散度

测试梯度只能确定两通量的散度相同。相差无散度场仍给相同弱方程，不应直接写成逐点通量相等。

9.2 Fixed Point Methods

9.2.1 Banach iteration and reaction–diffusion

Theorem 9.5 (Banach; source §9.2.1, Theorem 1)
If X is Banach and $A:X\to X$ satisfies $\|Au-Av\|\le q\|u-v\|$ for $0\le q<1$, it has a unique fixed point. For $u_{k+1}=Au_k$, $\|u_k-u\|\le q^k\|u_1-u_0\|/(1-q)$.

Expanded proof (completeness and an error estimate).

The successive differences are at most $q^k\|u_1-u_0\|$. Summing from k to $l-1$ shows that (u_k) is Cauchy with the stated tail bound. Completeness gives u, continuity gives $Au=u$, and two fixed points satisfy $\|u-v\|\le$

增长界不是导数界

线性增长不能单独给出 $|Da|\le M$。本稿在 H^2 命题中补明导数上界，以控制差商系数和误差项；存在性不需要它。

$q\|u-v\|$, hence coincide. The same proof works on any nonempty closed invariant subset of X. □

Let $F:\mathbb{R}^m\to\mathbb{R}^m$ be globally Lipschitz with constant L. Consider the zero-boundary system $u_t-\Delta u=F(u)$ with initial $g\in L^2(U;\mathbb{R}^m)$. A weak solution belongs to

$$u\in L^2(0,T;H_0^1),\qquad u_t\in L^2(0,T;H^{-1}),$$

obeys $\langle u_t,v\rangle+\int Du:Dv=\int F(u)\cdot v$ a.e., and has $u(0)=g$ in L^2. The time-continuity theorem gives $u\in C([0,T];L^2)$. The source starts with $g\in H_0^1$; the L^2 formulation below makes restart times unambiguous.

Theorem 9.6 (Reaction–diffusion; source Theorem 2)
There is a unique weak solution on every finite time interval. For two initial data its L^2 distance at time t is at most $e^{Lt}\|g-\tilde g\|_2$.

Expanded proof (short intervals with a uniform length).

On $X=C([0,h];L^2)$ let $Av=w$ solve the linear heat equation with forcing $F(v)$ and initial g. The linear theory of §7.1 applies since $\|F(v)\|_2\le L\|v\|_2+|F(0)||U|^{1/2}$. For $z=Av-A\tilde v$, zero initial data and the weak energy identity give

$$\tfrac12\frac{d}{dt}\|z\|_2^2+\|Dz\|_2^2\le LC_P\|v-\tilde v\|_2\|Dz\|_2\le\tfrac12\|Dz\|_2^2+\tfrac12L^2C_P^2\|v-\tilde v\|_2^2.$$

Thus $\|Av-A\tilde v\|_X\le LC_P\sqrt{h}\|v-\tilde v\|_X$. Choose h so this factor is less than one. It depends on L,U, not on g. Banach gives a solution on $[0,h]$; its endpoint is in L^2, so repeat the same argument on $[h,2h]$ and finitely many subsequent intervals. The matching L^2 traces eliminate time-interface jumps in the weak formulation. The finite sum of the interval energy norms remains finite.

For two solutions the difference satisfies $\tfrac12(\|z\|_2^2)'+\|Dz\|_2^2\le L\|z\|_2^2$. The Hilbert-triple chain rule justifies testing by z; Gronwall gives continuous dependence and uniqueness. □

全局 Lipschitz 的作用

短时间长度只依赖统一的 L，所以有限次拼接覆盖任意有限 T。对 $F(u)=u^2$，局部 Lipschitz 常数随振幅增长，不能用同一论证排除爆破。

9.2.2 Compact maps: Schauder and Schaefer

Theorem 9.7 (Schauder; source §9.2.2, Theorem 3)
If $K \subset X$ is nonempty, compact and convex and $A : K \to K$ is continuous, A has a fixed point.

Expanded proof (finite-dimensional approximation).

Cover K by finitely many balls $B(x_i, \varepsilon)$ with $x_i \in K$. Set $\rho_i(x) = (\varepsilon - \|x - x_i\|)_+$ and

$$P_\varepsilon x = \frac{\sum_i \rho_i(x) x_i}{\sum_i \rho_i(x)}.$$

The denominator is positive, P_ε is continuous, its range lies in $K_\varepsilon = \operatorname{conv}\{x_i\} \subset K$, and $\|P_\varepsilon x - x\| < \varepsilon$. The finite-dimensional convex-set form of Brouwer gives $x_\varepsilon = P_\varepsilon A x_\varepsilon$. Compactness provides a converging subsequence as $\varepsilon \to 0$; $\|x_\varepsilon - A x_\varepsilon\| < \varepsilon$ and continuity give $x = Ax$. □

A map is called compact here when it sends bounded sets to relatively compact sets. Continuity is an additional assumption.

Lemma 9.8 (A compact convex hull; editorial supplement)
In a Banach space the closed convex hull of a relatively compact set is compact.

Expanded proof (total boundedness survives convexification).

For a bounded set E with finite ε-net $\{x_i\}$, every convex combination of points of E is within ε of $\operatorname{conv}\{x_i\}$. The latter is compact, being the continuous image of a finite simplex, and has a finite ε-net. Thus the closed convex hull is totally bounded; it is complete as a closed subset of a Banach space, hence compact. □

Theorem 9.9 (Schaefer; source Theorem 4)
Let $A : X \to X$ be continuous and compact. If $\mathscr{E} = \{u : u = \lambda A u,\ 0 \le \lambda \le 1\}$ is bounded, then A has a fixed point.

Expanded proof (radial truncation and a strict radius).

Choose $R > \sup_{u \in \mathscr{E}} \|u\|$ and let $T(u) = Au \min\{1, R/\|Au\|\}$, taking $T(u) = 0$ if $Au = 0$. It is continuous and compact and maps $\overline{B}_R$ into itself. The preceding lemma makes $K = \overline{\operatorname{conv}} T(\overline{B}_R)$ compact; also $K \subset \overline{B}_R$ and $T(K) \subset K$. Schauder gives $u = T(u)$. If $\|Au\| > R$, then $u = \lambda A u$ with $\lambda = R/\|Au\| < 1$ and $\|u\| = R$, contradicting the strict choice of R. Therefore $u = Au$. The proof also works on a closed convex invariant set containing 0; closedness ensures its closed convex hull remains in the admissible set. □

9.2.3 An elliptic application with gradient dependence

Let $b : \mathbb{R}^n \to \mathbb{R}$ be globally Lipschitz, with constant L and $B = |b(0)|$. For $\mu > 0$ let $Av = w$ solve $-\Delta w + \mu w = -b(Dv)$ with zero trace, in $X = H_0^1(U)$.

Theorem 9.10 (Source Theorem 5, with an explicit sufficient bound)
If $\mu > L^2/2$, the problem $-\Delta u + b(Du) + \mu u = 0$, $u|_{\partial U} = 0$, has a solution $u \in H^2(U) \cap H_0^1(U)$.

Expanded proof (smoothing and a homotopy estimate).

Lax–Milgram and the H^2 estimate of §6.3 give

$$\|Av\|_{H^2} \le C_\mu (L\|Dv\|_2 + B|U|^{1/2}).$$

The Lipschitz bound on b gives continuity, even into H^2. Rellich's embedding $H^2(U) \Subset H^1(U)$ gives compactness in X. If $u = \lambda Au$ for $0 < \lambda \le 1$, then $-\Delta u + \mu u = -\lambda b(Du)$. Testing with u yields

$$\tfrac{1}{2}\|Du\|_2^2 + (\mu - L^2/2)\|u\|_2^2 \le B|U|^{1/2}\|u\|_2.$$

Writing $d = \mu - L^2/2 > 0$, Young bounds the right side by $d\|u\|_2^2/2 + B^2|U|/(2d)$. This uniformly bounds u in H_0^1; $\lambda = 0$ gives $u = 0$. Schaefer applies. The fixed point is in H^2 by its definition, so the PDE holds a.e. □

先验界必须沿同伦统一

只证明 $u = Au$ 的解有界还不够。截断固定点可能满足 $u = \lambda Au$；需对所有 $\lambda \in [0,1]$ 使用同一个界，才能排除它落在截断球面。

9.3 Method of Subsolutions and Supersolutions

9.3.1 Weak order and boundary traces

Let $f : \mathbb{R} \to \mathbb{R}$ be globally Lipschitz, with constant L. For $\underline{u}, \overline{u} \in H^1(U)$, call them a subsolution and a supersolution of $-\Delta u = f(u)$ when, for all nonnegative $v \in H_0^1(U)$,

$$\int D\underline{u} \cdot Dv \le \int f(\underline{u})v, \qquad \int D\overline{u} \cdot Dv \ge \int f(\overline{u})v.$$

The boundary conditions required below are inequalities of Sobolev traces, not pointwise values of arbitrary representatives.

存在性与迭代收敛

Schauder、Schaefer 给出固定点存在性，不自动保证 $u_{k+1} = Au_k$ 收敛。要证明这条序列收敛，还须另证压缩性、单调性或其他稳定结构。

Theorem 9.11 (Ordered existence; source Theorem 1)
If $\underline{u} \le \overline{u}$ a.e. and $\underline{u}|_{\partial U} \le 0 \le \overline{u}|_{\partial U}$, there is a weak solution $u \in H_0^1(U)$ between them. The increasing and decreasing iterations give respectively the least and greatest weak solution in this order interval.

Expanded proof (comparison, domination, and strong convergence).

Choose $\lambda \ge L$, $\lambda > 0$, so $h(s) = f(s) + \lambda s$ is nondecreasing. Starting from $u_0 = \underline{u}$, solve

$$(-\Delta + \lambda)u_{k+1} = h(u_k), \qquad u_{k+1} \in H_0^1(U). \tag{9.3}$$

The forcing is in L^2 and Lax–Milgram gives every iterate. Subtract the equation for u_1 from the subsolution inequality after adding the same $\lambda\underline{u}$ term. Test by $v = (\underline{u} - u_1)_+$. The trace inequality makes $v \in H_0^1$, and the Sobolev truncation rule gives $\|Dv\|_2^2 + \lambda\|v\|_2^2 \le 0$. Hence $u_0 \le u_1$.

If $u_{k-1} \le u_k$, subtract successive equations and test by $(u_k - u_{k+1})_+$. Since $h(u_{k-1}) \le h(u_k)$, the same reasoning gives $u_k \le u_{k+1}$. Comparing with the supersolution using $(u_{k+1} - \overline{u})_+$ gives $u_{k+1} \le \overline{u}$ whenever $u_k \le \overline{u}$. Thus

$$\underline{u} = u_0 \le u_1 \le \cdots \le u_k \le \overline{u}.$$

The pointwise limit u exists a.e. Also $|u_k| \le |\underline{u}| + |\overline{u}| \in L^2$; dominated convergence gives $u_k \to u$ in L^2. The Lipschitz property gives $h(u_k) \to h(u)$ in L^2. The bounded inverse $(-\Delta + \lambda)^{-1} : L^2 \to H_0^1$ now shows that $u_{k+1} \to (-\Delta + \lambda)^{-1}h(u)$ strongly in H_0^1. Its L^2 limit was u, hence u is this inverse image and solves the nonlinear equation after cancellation of λu.

If v is any solution in the interval, induction by the same comparison gives $u_k \le v$, hence $u \le v$. Starting instead at $\overline{u}$ gives a decreasing sequence and the greatest solution. These extremal properties are editorial consequences; uniqueness is not asserted. □

正部测试的合法性

当边界迹满足 $a \le b$ 时，$(a-b)_+$ 的迹为零，因此属于 H_0^1。没有边界序，积分中的非负项不能用来推出全域比较。

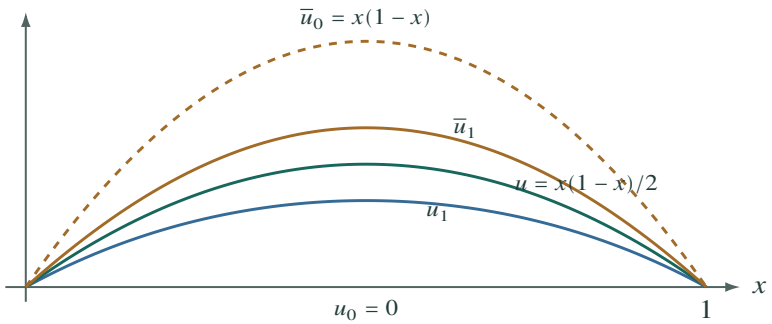


Diagram T9.2. Two ordered iterations trap the solution. For $-u'' = 1$ on $(0,1)$ with zero endpoint data, take $u_0 = 0$, $\overline{u}_0 = x(1-x)$ and shift $\lambda = 4$. The curves show the exact first lower and upper iterates and the solution $u = x(1-x)/2$. Later lower iterates increase and upper iterates decrease in the pointwise order.

9.3.2 Local nonlinearities and the limits of comparison

If the barriers are bounded and f is only Lipschitz on their common value interval $[a,b]$, extend $f|_{[a,b]}$ to a globally Lipschitz function (for example by constant continuation outside). The iteration stays between the barriers, so the resulting equation uses the original f. This is a useful editorial localization of the source's hypothesis.

If f is nonincreasing, two solutions with the same trace are identical: testing their difference gives $\|D(u-v)\|_2^2 = \int (f(u)-f(v))(u-v) \le 0$. More generally uniqueness follows when $\mathrm{Lip}(f)C_P^2 < 1$. This uses a quantitative bound; mere smoothness is insufficient.

上下解不保证唯一

对 $-\Delta u = \lambda_1 u$，零解和任意倍数的第一特征函数都是解。顺序构造给出区间内极小、极大解；唯一性需要额外结构。

9.4 Nonexistence of Solutions

9.4.1 Large data: a weighted moment becomes infinite

Assume U is connected and let $\phi_1 > 0$ be the first Dirichlet eigenfunction, $-\Delta\phi_1 = \lambda_1\phi_1$, normalized by $\int_U \phi_1 = 1$. For the problem $u_t - \Delta u = u^2$, $u|_{\partial U} = 0$, $u(0) = g \ge 0$, take a classical solution with enough boundary regularity for Green's identity. The maximum principle gives $u \ge 0$ as long as it exists.

Theorem 9.12 (Large-data obstruction; source §9.4.1, Theorem 1)
If $\eta_0 = \int g\phi_1 > \lambda_1$, a classical solution cannot persist through the time

$$T_* = \frac{1}{\lambda_1} \log \frac{\eta_0}{\eta_0 - \lambda_1}.$$

This is an upper bound for its lifespan, not necessarily its actual blow-up time.

Expanded proof (normalization gives Jensen's inequality).

Set $\eta(t) = \int u(t)\phi_1$. Both u and ϕ_1 vanish on the boundary, so the two boundary terms in Green's second identity vanish. Consequently

$$\eta' = -\lambda_1 \eta + \int u^2 \phi_1 \geq -\lambda_1 \eta + \eta^2.$$

The last step uses the probability measure $\phi_1 dx$. Comparison with the scalar ODE, or integration of $\xi = e^{\lambda_1 t}\eta$, gives

$$\eta(t) \geq \frac{\lambda_1 \eta_0 e^{-\lambda_1 t}}{\lambda_1 - \eta_0(1 - e^{-\lambda_1 t})} \qquad (0 \leq t < T_*).$$

The denominator stays positive until T_* and tends to zero there. If the PDE solution existed smoothly through that time, its finite moment would contradict this bound. It might cease to exist earlier. Moreover $\eta(t) \leq \|u(t)\|_\infty$, so moment divergence implies amplitude divergence when the solution persists up to the comparison time. □

9.4.2 Subcritical Fujita blow-up in all of space

Theorem 9.13 (Small-data obstruction; source Theorem 2)
Let $g \in C_c^\infty(\mathbb{R}^n)$ be nonnegative and nonzero. If $1 < p < 1 + 2/n$, there is no global nonnegative classical solution of $u_t - \Delta u = u^p$ in the class $C([0,T]; L^1 \cap L^\infty)$ for every finite T. This class supplies the decay and integrability for the calculation.

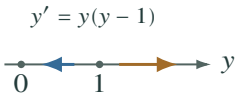
Expanded proof (a scale selected after the initial datum).

Fix $s > 0$, put $\Phi_s(x) = (4\pi s)^{-n/2} e^{-|x|^2/(4s)}$, and set $\eta(t) = \int u(t)\Phi_s$. Since

$$\Delta \Phi_s = \left(\frac{|x|^2}{4s^2} - \frac{n}{2s} \right) \Phi_s, \qquad \int \Phi_s = 1,$$

Green's identity and Jensen give $\eta' \geq -\lambda\eta + \eta^p$ with $\lambda = n/(2s)$. To justify the identity on $\mathbb{R}^n$, test first with $\chi_R \Phi_s$. After integrating by parts onto this compactly supported test, the errors $u(2D\chi_R \cdot D\Phi_s + \Phi_s \Delta \chi_R)$ tend to zero. On each finite time interval, the assumed L^1, L^∞ bounds also

图 T9.M：反应与扩散的阈值



归一化为 $\lambda_1 = 1$ 后，比较 ODE 为 $y' = y(y-1)$。箭头只表示这个 ODE 的相线；$y_0 \leq 1$ 时不能据此断言 PDE 全局存在。

control u^p; dominated convergence gives the integral differential identity. Thus no unproved decay of Du is required.

The comparison ODE blows up if $\eta(0)^{p-1} > \lambda$. This condition is equivalent to

$$(4\pi)^{-n/2} \int g(x) e^{-|x|^2/(4s)} dx > (n/2)^{1/(p-1)} s^{n/2-1/(p-1)}.$$

As $s \to \infty$, the left side tends to a positive constant and the right side tends to zero, since $p < 1 + 2/n$. Choose one such finite s; the ODE comparison then forbids a global solution. At $p = 1 + 2/n$ the right side is constant: this argument does not establish the endpoint. □

9.4.3 The full Derrick–Pohozaev identity

Let $n \geq 3, f \in C(\mathbb{R})$, $F(s) = \int_0^s f(r)dr$, and $u \in C^2(\overline{U})$ solve $-\Delta u = f(u)$ with zero boundary data.

Theorem 9.14 (Identity and nonexistence; source §9.4.2, Theorem 3)
One has

$$\frac{n-2}{2} \int_U |Du|^2 + \frac{1}{2} \int_{\partial U} (x \cdot \nu)(\partial_\nu u)^2 = n \int_U F(u). \tag{9.4}$$

If U is star-shaped about 0 and $f(u) = |u|^{p-1}u$ with $p > (n+2)/(n-2)$, then $u \equiv 0$.

Expanded proof (every boundary term retained).

Introduce the vector field $Q = (x \cdot Du)Du - x|Du|^2/2$. Direct differentiation cancels the Hessian cross terms and yields

$$\operatorname{div} Q = (x \cdot Du)\Delta u + \frac{2-n}{2}|Du|^2.$$

On ∂U, all tangential derivatives of u vanish, so $Du = (\partial_\nu u)\nu$ and $Q \cdot \nu = (x \cdot \nu)(\partial_\nu u)^2/2$. Also $f(u)x \cdot Du = x \cdot DF(u)$, and $\int x \cdot DF(u) = -n \int F(u)$ since $F(0) = 0$ on the boundary. The divergence theorem gives (9.4) with the displayed signs.

For a C^1 star-shaped boundary, choose a local defining function ρ with $U = \{\rho < 0\}$. Since $(1-t)x \in \overline{U}$ for small $t > 0$, $\rho((1-t)x) \leq 0 = \rho(x)$, so $-D\rho(x) \cdot x \leq 0$; hence $x \cdot \nu \geq 0$. Testing the PDE by u gives $\int |Du|^2 = \int |u|^{p+1}$. Substitution in the identity implies

$$\left(\frac{n-2}{2} - \frac{n}{p+1}\right) \int |u|^{p+1} \leq 0.$$

The coefficient is strictly positive at a supercritical exponent, so $u = 0$. At equality of the exponents it is zero: the displayed inequality alone

两个临界指数

热方程这里出现 $1 + 2/n$，来自质量与扩散尺度；椭圆 Pohozaev 中出现 $(n+2)/(n-2)$，来自 Dirichlet 能量与幂次能量的尺度。两者不能混同。

gives no conclusion. On a strictly star-shaped domain a positive nonzero solution at the critical exponent is also ruled out by the boundary term and Hopf's lemma, but this is a separate argument and does not extend the stated theorem to arbitrary sign-changing critical solutions. □

9.5 Geometric Properties of Solutions

9.5.1 Dilation derivatives and star-shaped level sets

Let $V \Subset W$ be smooth bounded star-shaped domains about 0, with $0 \in V$, and assume $U = W \setminus \overline{V}$ is connected. Suppose u is harmonic in U, smooth up to the boundary, equal to 1 on ∂V and 0 on ∂W.

Theorem 9.15 (Source §9.5.1, Theorem 1)
For $0 < \ell < 1$, the level surface $\{u = \ell\}$ is smooth and bounds a star-shaped set. In fact $x \cdot Du < 0$ throughout U.

Expanded proof (the sign on the inner boundary).

The maximum principle gives $0 < u < 1$. Set $v = x \cdot Du$. The commutator formula $\Delta(x \cdot Du) = 2\Delta u + x \cdot D\Delta u$ shows that v is harmonic. On the outer boundary Du is a negative multiple of the outward normal of W, and $x \cdot \nu_W \geq 0$, so $v \leq 0$. On the inner boundary Du points into V, hence is a negative multiple of its outward normal ν_V; once more $v \leq 0$. The strong maximum principle gives either $v < 0$ or $v \equiv 0$ in the connected shell. The latter is impossible: on each ray joining the two boundaries it would make u constant, contrary to the values $1, 0$. Thus $v < 0$, so $Du \neq 0$ in U and the implicit function theorem makes every intermediate level smooth.

Extend $u = 1$ inside V. On a ray $r\omega$, star-shapedness of the two domains makes their intersections initial radial intervals. In the shell, $\partial_r u(r\omega) = v(r\omega)/r < 0$. Therefore each ray crosses an intermediate level exactly once, and its superlevel set contains every segment from 0 to one of its points. This proves star-shapedness directly, including the global radial geometry. □

9.5.2 Maximum principles on a small set

For moving planes the zero-order coefficient need not have a sign. We record two precise substitutes for an unrestricted maximum principle.

边界几何进入哪一步

星形性只用来保证 $x \cdot \nu \geq 0$。Pohozaev 恒等式本身对一般光滑区域成立；若区域不是星形，不能直接丢掉边界积分。

内边界法向易混

相对于壳域 U，内边界外法向指向 V；相对于 V 则方向相反。证明中用 ν_V，于是 Du 为其负倍数，两层边界都给 $x \cdot Du \leq 0$。

Lemma 9.16 (Nonnegative supersolutions; source §9.5.2, Lemma 1)
Let D be connected, $c \in L^\infty(D)$, and $v \geq 0$, $v \not\equiv 0$ satisfy $-\Delta v + cv \geq 0$ classically. Then $v > 0$ in D. At a zero boundary point with an interior tangent ball and C^1 regularity, $\partial_\nu v < 0$.

Expanded proof (remove the zero-order term).

Put $v = e^{kx_1} w$ with $k^2 \geq \|c\|_\infty$. Then $w \geq 0$ and $-\Delta w - 2k w_{x_1} \geq (k^2 - c) w \geq 0$. The strong maximum principle and Hopf's lemma for this operator, whose zero-order coefficient is zero, apply. At a zero of v, $\partial_\nu w = e^{-kx_1} \partial_\nu v$, giving the asserted sign. □

Lemma 9.17 (Small-support comparison; editorial supplement)
Fix a ball B and $L \geq 0$. There is $\delta > 0$ such that if $D \subset B$, $w_- \in H_0^1(D)$, $-\Delta w = c(x) w$, $|c| \leq L$, and $|\{w < 0\}| < \delta$, then $w \geq 0$.

Expanded proof (the quantitative continuation step).

Choose $q > 2$ for which $\|z\|_{L^q(B)} \leq C_B \|Dz\|_2$ on $H_0^1(B)$ (take $q = 2n/(n-2)$ if $n \geq 3$, and $q = 4$ if $n = 1, 2$). Extend w_- by zero. Testing the equation with $-w_-$ gives

$$\|Dw_-\|_2^2 \leq L \|w_-\|_2^2 \leq L C_B^2 |\{w < 0\}|^{1-2/q} \|Dw_-\|_2^2.$$

Choose $L C_B^2 \delta^{1-2/q} < 1$ and absorb. Poincaré gives $w_- = 0$. The constant depends on the containing ball, not on the shape of D. □

9.5.3 Moving planes and radial symmetry

Let $B = B_1(0)$, let f be Lipschitz on $[0, \max u]$, and suppose $u \in C^2(\overline{B})$, $-\Delta u = f(u)$, $u > 0$ in B, $u = 0$ on ∂B. For $0 \leq \lambda < 1$, set

$$\Sigma_\lambda = B \cap \{x_n > \lambda\}, \quad x^\lambda = (x', 2\lambda - x_n), \quad w_\lambda(x) = u(x^\lambda) - u(x).$$

Reflection keeps the cap inside the ball. The difference satisfies $-\Delta w_\lambda = c_\lambda w_\lambda$ with $|c_\lambda| \leq L$, using the difference quotient of f (set it to zero where the two values agree). On the plane $w_\lambda = 0$; on the spherical cap it is nonnegative, and strictly positive away from the rim when $\lambda > 0$.

小区域替代系数符号

比较差满足 $-\Delta w = cw$，c 可正可负。负部支集足够小时，Sobolev 与 Hölder 给出小因子，才能吸收 $L\|w_-\|_2^2$；普通最大值原理不能直接套用。

Theorem 9.18 (Radial symmetry; source Theorem 2)
There is a radial profile $u(x) = v(|x|)$ and $v'(r) < 0$ for $0 < r < 1$.

Expanded proof (start, stop, and continue the plane).

For λ close to 1 the cap has measure less than δ from the small-support lemma. Its boundary values imply $(w_\lambda)_- \in H_0^1(\Sigma_\lambda)$; the lemma gives $w_\lambda \geq 0$. The previous strong principle gives $w_\lambda > 0$. Let λ_0 be the infimum of positions such that all planes further out have this comparison. Continuity gives $w_{\lambda_0} \geq 0$.

If $\lambda_0 > 0$, the spherical boundary values exclude $w_{\lambda_0} \equiv 0$, hence $w_{\lambda_0} > 0$ in the cap. Choose a compact set $K \Subset \Sigma_{\lambda_0}$ whose complement in the cap has measure less than $\delta/2$. On K the minimum of w_{λ_0} is positive. Uniform continuity therefore gives $w_\lambda > 0$ on K for all λ slightly below λ_0. The extra thin slice of the cap has measure less than $\delta/2$. Thus any negative set of w_λ has measure less than δ; the small-support lemma excludes it. The strong principle restores strict positivity, contradicting the stopping definition. Hence $\lambda_0 = 0$.

At $\lambda = 0$ one inequality follows. Repeating from the opposite pole gives the reverse inequality, so u is symmetric in the central plane. Rotate the argument to every direction: invariance under all reflections through 0 makes u radial. For $0 < \lambda < 1$, Hopf at an interior point of the flat face gives $\partial_\nu w_\lambda < 0$ with $\nu = -e_n$. Since $(w_\lambda)_{x_n} = -2u_{x_n}$ there, $u_{x_n} < 0$ on that face. Taking $x = re_n$, $\lambda = r$, proves $v'(r) < 0$. At the center the gradient is zero by symmetry. This proof replaces the source's boundary-derivative continuation with a fully quantified small-support argument; it covers the same positive ball problem. □

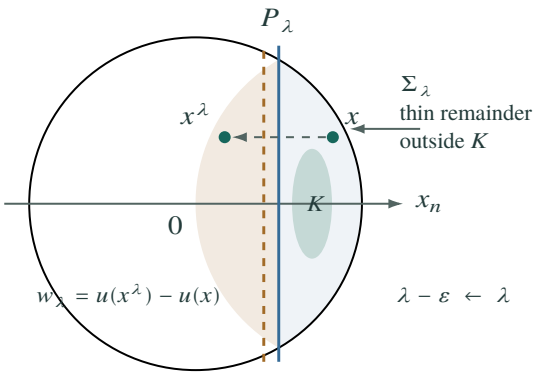


Diagram T9.3. Reflection and the region available for continuation. The cap Σ_λ reflects inside the unit ball. At a proposed positive stopping position, comparison is uniformly strict on a compact core K. Any negative part for a nearby plane must lie in the thin remainder, where the small-support estimate excludes it. Geometry is schematic; the equation is scalar and $u > 0$ is essential.

9.6 Gradient Flows

9.6.1 Extended energies and subgradients

Let H be a real Hilbert space and $I : H \to (-\infty, +\infty]$ be proper, convex and lower semicontinuous in the norm topology. Its effective domain is $D(I) = \{u : I(u) < \infty\}$. For $u \in D(I)$ set

$$\partial I(u) = \{a \in H : I(w) \geq I(u) + (a, w - u) \text{ for every } w \in H\};$$

outside $D(I)$ the set is empty. Write $A = \partial I$ and $D(A) = \{u : A(u) \neq \emptyset\}$. Adding the subgradient inequalities shows $(a - b, u - v) \geq 0$ for $a \in A(u)$, $b \in A(v)$. Also $0 \in A(u)$ exactly when u minimizes I.

Theorem 9.19 (Resolvent; source §9.6.1, Theorem 1)
For each $\lambda > 0$ and $w \in H$ there is exactly one u with $u + \lambda A(u) \ni w$. It uniquely minimizes

$$I(z) + \frac{1}{2\lambda}\|z - w\|^2.$$

Expanded proof (coercivity and the actual subgradient inequality).

The closed convex epigraph of I can be strictly separated from a point $(u_0, I(u_0) - 1)$ below it, where $I(u_0) < \infty$. The separating functional has a nonzero vertical component: otherwise its values at these two points with the same first coordinate could not be strictly ordered. Its sign follows from upward closure of the epigraph. Dividing gives an affine minorant $I(z) \geq (b, z) + d$. This uses the geometric Hahn–Banach theorem as an explicit functional analysis input. Therefore the quadratic functional above is coercive and bounded below. Norm-closed convex sublevel sets are weakly closed by the same separation theorem, so I is weakly lower semicontinuous. A bounded minimizing sequence has a weakly convergent subsequence in H; the limit minimizes. The quadratic term makes the minimizer unique.

Let that minimizer be u. Compare it with $u + t(z - u)$ for $0 < t < 1$. Convexity bounds its energy above by $(1 - t)I(u) + tI(z)$. Expansion of the quadratic, division by t, and $t \downarrow 0$ give $I(z) - I(u) \geq ((w - u)/\lambda, z - u)$. Thus $(w - u)/\lambda \in A(u)$. Conversely this inequality, together with expansion of the square, proves minimality, with a strict excess $\|z - u\|^2/(2\lambda)$ if $z \neq u$. □

Define the resolvent $J_\lambda = (\mathrm{Id} + \lambda A)^{-1}$ and the Yosida approximation $A_\lambda = (\mathrm{Id} - J_\lambda)/\lambda$.

正性和球形的用途

球形保证反射帽仍在定义域内；正性给球面边界上的严格比较，从而排除正停止位置。一般区域或变号解不能据此推出径向对称。

允许无穷能量

把不满足边界或约束的函数赋值 $+\infty$，能把限制条件写进定义域。$\partial I(u)$ 是 Hilbert 空间中的向量集合，依赖所选内积，不能只看形式变分。

Theorem 9.20 (Source Theorem 2, with sharpened Lipschitz bound)
For $u, v \in H$, writing $j = J_\lambda u - J_\lambda v$ and $a = A_\lambda u - A_\lambda v$, one has

$$\|j\|^2 + \lambda^2 \|a\|^2 \le \|u - v\|^2, \qquad (a, u - v) \ge 0.$$

In particular J_λ is nonexpansive, A_λ is $1/\lambda$- Lipschitz, and $A_\lambda u \in A(J_\lambda u)$. For $u \in D(A)$ and $a_0 \in A(u)$, $\|A_\lambda u\| \le \|a_0\|$ and $\|J_\lambda u - u\| \le \lambda \|a_0\|$. The last convergence extends to $u \in \overline{D(A)}$.

Expanded proof (firm nonexpansiveness).

The resolvent identity gives $u - v = j + \lambda a$, with $(a, j) \ge 0$. Squaring proves the first inequality; taking inner product with a proves monotonicity of A_λ. The inclusion is the defining resolvent equation. Comparing $(J_\lambda u, A_\lambda u)$ with (u, a_0) by monotonicity gives $\lambda \|A_\lambda u\|^2 \le \lambda (a_0, A_\lambda u)$, which gives the bound and convergence on $D(A)$. Approximate any point of its closure by $v \in D(A)$ and use $\|J_\lambda u - u\| \le 2\|u - v\| + \|J_\lambda v - v\|$. The source states the weaker $2/\lambda$ Lipschitz estimate; the squared identity supplies the sharper bound used here. □

Lemma 9.21 (Maximality and graph closure; editorial supplement)
If (u, a) satisfies $(a - b, u - v) \ge 0$ for every $b \in A(v)$, then $a \in A(u)$. Consequently the graph is closed for strong convergence of u and weak convergence of bounded a.

Expanded proof (one resolvent test).

Take $v = J_1(u + a)$ and $b = u + a - v \in A(v)$. The inequality becomes $-\|u - v\|^2 \ge 0$, so $u = v$, $a = b$. For a strong–weak limit of graph pairs, monotonicity passes to the limit against every fixed pair (v, b), and the first assertion applies. □

9.6.2 Constructing the nonlinear flow

Theorem 9.22 (Gradient flow; source §9.6.2, Theorem 3)
For $g \in D(A)$ there is a unique $u \in C([0, \infty); H)$, $u' \in L^\infty(0, \infty; H)$ such that $u(0) = g$, $u(t) \in D(A)$ for every $t \ge 0$, and $-u'(t) \in A(u(t))$ a.e. For any $a_0 \in A(g)$, $\|u'\|_{L^\infty} \le \|a_0\|$. If $\overline{D(A)} = H$, these solutions extend uniquely by contraction to a strongly continuous nonlinear semigroup $S(t)$ on H. For general $g \in H$ this extension initially asserts a continuous semigroup solution, without the global derivative bound above.

图闭性不是弱连续性

这里用“输入强收敛、输出弱收敛”和单调图的最大性。并没有假设多值算子 A 连续，也不能把 $A(u_k)$ 直接取极限写为 $A(u)$。

Expanded proof (Yosida approximation and a common graph test).

For $\lambda > 0$ solve the globally Lipschitz Hilbert-space ODE $u_\lambda' = -A_\lambda u_\lambda, u_\lambda(0) = g$. Picard iteration gives a global solution (linear growth prevents finite-time escape). Monotonicity gives contraction between two such trajectories. Compare one trajectory with its time translate, divide by the shift and let the shift vanish to obtain

$$\|u_\lambda'(t)\| \le \|A_\lambda g\| \le M := \|a_0\|.$$

Write $a_\lambda = A_\lambda u_\lambda$, $z_\lambda = J_\lambda u_\lambda$ so $u_\lambda = z_\lambda + \lambda a_\lambda$ and $a_\lambda \in A(z_\lambda)$. For two parameters, monotonicity gives

$$\tfrac{1}{2}(\|u_\lambda - u_\mu\|^2)' \le -(a_\lambda - a_\mu, \lambda a_\lambda - \mu a_\mu) \le 2(\lambda + \mu)M^2.$$

Thus the trajectories are Cauchy uniformly on every $[0,T]$, with $\|u_\lambda - u_\mu\|^2_{C([0,T];H)} \le 4T(\lambda + \mu)M^2$. They converge strongly to a Lipschitz u; after extracting on finite intervals, $a_\lambda \rightharpoonup a = -u'$ in $L^2(0,T;H)$. Also $z_\lambda \to u$ uniformly and $\|a\|_{L^\infty} \le M$.

To identify the graph without exceptional sets depending on arbitrary test vectors, define the measurable pair

$$v(t) = J_1(u(t) + a(t)), \qquad b(t) = u(t) + a(t) - v(t) \in A(v(t)).$$

Both functions are in $L^2(0,T;H)$ since the resolvent is Lipschitz. Integrate $(a_\lambda - b, z_\lambda - v) \ge 0$ and pass to the limit using strong convergence of z_λ and weak convergence of a_λ. The result is $\int (a - b, u - v) = -\int \|u - v\|^2 \ge 0$. Hence $u = v$ a.e., and $a = b \in A(u)$ a.e. This is the essential limit identification; pointwise convergence of a_λ was not assumed.

For any fixed t, choose times $t_j \to t$ outside the null set so $(u(t_j), a(t_j))$ are graph pairs and $\|a(t_j)\| \le M$. A weakly convergent subsequence of the second components and the strong–weak graph closure give $u(t) \in D(A)$. Two solutions satisfy $\frac{1}{2}(\|u - v\|^2)' \le 0$ by monotonicity, proving uniqueness and contraction. Uniqueness also gives $S(t + s) = S(t)S(s)$ on $D(A)$.

For $g \in \overline{D(A)}$, approximate by $g_j \in D(A)$. Contraction makes $S(t)g_j$ Cauchy uniformly even on all $t \ge 0$. The limit is independent of the approximating sequence, continuous at $t = 0$, and retains contraction and the semigroup law. It need not have a derivative uniformly bounded down to $t = 0$. This distinguishes the strong solution theorem from its continuous extension. □

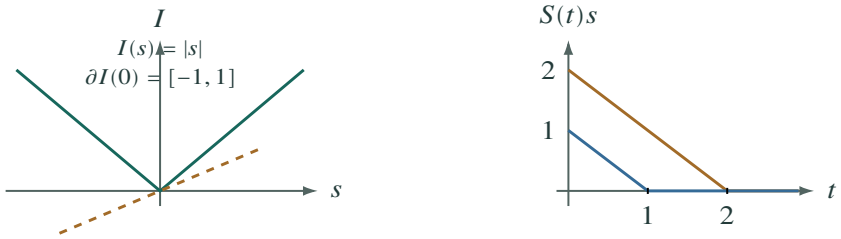


Diagram T9.4. A nonsmooth energy can merge trajectories. For $H = \mathbb{R}$ and $I(s) = |s|$, $J_\lambda(s) = \operatorname{sgn}(s)(|s| - \lambda)_+$ and $S(t)s = \operatorname{sgn}(s)(|s| - t)_+$. The supporting slopes at zero form $[-1, 1]$; different positive initial values reach zero at different times and thereafter coincide. These are exact graphs, not a discretization of another model.

9.6.3 Dissipation and a quasilinear parabolic equation

Lemma 9.23 (Convex chain rule; editorial supplement)
If $u \in W^{1,2}(0,T;H)$ and $a \in L^2(0,T;H)$ with $a(t) \in \partial I(u(t))$ a.e., then $I(u(t))$ has an absolutely continuous representative and $(I(u))' = (a, u')$ a.e. For the flow,

$$I(u(t)) + \int_s^t \|u'(r)\|^2 dr = I(u(s)).$$

For the solution above with $g \in D(A)$ the identity holds for all $0 \le s \le t$.

Expanded proof (two subgradient inequalities squeeze the derivative).

At times where the inclusions hold, for $h > 0$ the increment of I between $u(t)$ and $u(t+h)$ lies between $(a(t), u(t+h) - u(t))$ and $(a(t+h), u(t+h) - u(t))$. On compact subintervals these bounds also make $I(u)$ integrable: use one fixed graph point as an affine lower bound, and test the subgradient at $u(t)$ against a fixed finite-energy point for an upper bound in L^1. Multiply the increment inequalities by a nonnegative smooth test in time and integrate. Difference quotients of u converge to u' in L^2 and translations of a converge to a in L^2. Both bounds therefore converge to (a, u') in distributions. This L^1 derivative gives the asserted absolutely continuous representative. For our flow a is bounded. The subgradient inequalities and continuity of u then identify this representative with the actual $I(u(t))$ at every time, including 0, by graph closure and lower semicontinuity. Finally $a = -u'$ gives dissipation. □

Let $L \in C^2(\mathbb{R}^n)$ satisfy $\theta|\xi|^2 \le D^2L(p)\xi \cdot \xi$ and $|D^2L(p)| \le M$. On $H = L^2(U)$ define $I(u) = \int_U L(Du)$ for $u \in H_0^1(U)$, and $I(u) = +\infty$ otherwise.

收缩不代表严格收缩

这里 $\|S(t)g - S(t)h\| \le \|g - h\|$，因子是 1，不是 Banach 定理所需的小于 1。半群还可能把不同初值合并，见习题 11。

Theorem 9.24 (PDE realization; source §9.6.3, Theorem 4)
This energy is proper, convex and lower semicontinuous in L^2. Its subdifferential is single-valued with

$$D(\partial I) = H^2(U) \cap H_0^1(U), \qquad \partial I(u) = -\operatorname{div} DL(Du).$$

For g in this domain the flow solves $u_t - \operatorname{div} DL(Du) = 0$, with zero boundary data, and $u \in L^\infty(0,\infty; H^2 \cap H_0^1)$, $u_t \in L^\infty(0,\infty; L^2)$. For $g \in L^2$ its continuous semigroup extension is defined by approximation.

Expanded proof (the topology of the energy matters).

Taylor's inequality and $\int_U Du = 0$ for $u \in H_0^1$ give $I(u) \geq L(0)|U| + \theta\|Du\|_2^2/2$; the affine term integrates to zero by density of C_c^∞. A sequence converging in L^2 with bounded energy is thus bounded in H_0^1; a subsequence converges weakly there to the same limit. Convex integral lower semicontinuity from §8.2 gives the L^2 lower semicontinuity, including the case of infinite limit energy. Properness follows from $I(0) < \infty$.

If $u \in H^2 \cap H_0^1$, the Lipschitz bound on DL makes $-\operatorname{div} DL(Du) \in L^2$. The convex supporting inequality, integrated and integrated by parts against $w - u \in H_0^1$, shows it belongs to $\partial I(u)$; tests outside H_0^1 have infinite energy. Conversely, if $a \in \partial I(u)$, use $w = u \pm t\varphi$ for $\varphi \in H_0^1$. The linear growth of DL justifies differentiation under the integral, giving $\int a\varphi = \int DL(Du) \cdot D\varphi$. The H^2 result of §9.1 applies to DL, whose derivative is bounded and uniformly elliptic. Therefore $u \in H^2 \cap H_0^1$, and density of test functions identifies a uniquely. The same estimates, testing first by u, give $\|u\|_{H^2} \leq C\|a\|_2$; the constant vector $DL(0)$ has zero pairing with zero-trace gradients.

The domain is dense in L^2 because it contains $C_c^\infty(U)$. Apply the flow theorem; its speed bound and this elliptic estimate give the claimed uniform H^2 bound. In weak form the spatial divergence is exactly the L^2 subgradient identified above. The statement for rough g uses contraction extension, not the uniform-in-time H^2 claim made for domain data. □

Hilbert 梯度依赖度量

这里取 $H = L^2$，所以梯度为 $-\operatorname{div} DL(Du)$。若在 H_0^1 的梯度内积中取 Riesz 代表，还需再作用逆 Laplace 算子；所得流不同。

9.7 Problems

The following short problem descriptions preserve the source numbers 1–14. All solutions are editorial additions. Domain, decay, and regularity conditions needed by a computation are stated explicitly.

1. A flow proof of the finite-dimensional zero lemma

Solution. Let V be smooth near $\overline{B}_r$ and solve $x' = -V(x)$. The inequality $V(x) \cdot x \geq 0$ on the sphere makes the closed ball forward invariant. One justification uses a Lipschitz bound near the ball: for $d(x) = \operatorname{dist}(x, \overline{B}_r)$, projection to the sphere gives $(d(x)^2)' \leq Cd(x)^2$ outside, so an orbit starting with $d = 0$ cannot leave by Gronwall. Boundedness in the compact ball permits global continuation. For every $h > 0$ its continuous time-h map has a Brouwer fixed point y_h. Take $h_j \downarrow 0$ and $y_{h_j} \to y$. Uniformly for initial points in the ball, $x(h, y_h) = y_h - hV(y_h) + O(h^2)$, since V and DV are bounded there. The fixed-point identity gives $V(y_h) = O(h)$, hence $V(y) = 0$. A fixed point of just one time map could be a periodic orbit; the small-time limit is necessary to complete this argument. □

2. Weakly continuous scalar nonlinearities are affine

Solution. Write G for the continuous scalar nonlinearity. Fix $s, t \in \mathbb{R}$, $0 < \theta < 1$. On each interval of length $1/k$ let f_k equal s on the first fraction θ and t on the remainder. For continuous test functions, cellwise averaging gives $f_k \rightharpoonup m = \theta s + (1-\theta)t$; boundedness and density extend this to all L^2 tests. The same argument gives $G(f_k) \rightharpoonup \theta G(s) + (1-\theta)G(t)$. The assumed weak continuity forces $G(\theta s + (1-\theta)t) = \theta G(s) + (1-\theta)G(t)$. A continuous function satisfying this identity on every segment is $G(s) = \alpha s + \beta$: set $\beta = G(0)$, compare first on $[0,1]$, then use segments containing $0, 1$ and any real s. Conversely affine maps preserve weak convergence by linearity. □

3. Penalty approximation of the nonnegative obstacle

Solution. Let $\beta_\varepsilon(s) = \min(s, 0)/\varepsilon$ and solve $-\Delta u_\varepsilon + \beta_\varepsilon(u_\varepsilon) = f$. Existence follows by minimizing the strictly convex functional

$$\tfrac{1}{2}\|Dv\|_2^2 + \frac{1}{2\varepsilon}\|v_-\|_2^2 - \int fv \quad \text{on } H_0^1(U).$$

Coercivity and convex lower semicontinuity give a minimizer, and its derivative gives the equation. Testing by u_ε bounds $\|Du_\varepsilon\|_2$ and $\varepsilon^{-1}\|(u_\varepsilon)_-\|_2^2$. Testing by $-(u_\varepsilon)_-$ gives more precisely

$$\|D(u_\varepsilon)_-\|_2^2 + \varepsilon^{-1}\|(u_\varepsilon)_-\|_2^2 \leq \|f_-\|_2 \|(u_\varepsilon)_-\|_2,$$

so $\|(u_\varepsilon)_-\|_2 \leq \varepsilon\|f_-\|_2$. A weakly convergent subsequence has limit $u \geq 0$ by compact L^2 convergence. For each $w \geq 0$ in H_0^1, $\beta_\varepsilon(u_\varepsilon)(w - u_\varepsilon) \leq 0$, hence

$$\int Du_\varepsilon \cdot D(w - u_\varepsilon) \geq \int f(w - u_\varepsilon).$$

Rearrange with $\|Du_\varepsilon\|_2^2$ on the left and use weak lower semicontinuity to obtain the desired variational inequality for u. Two such solutions, tested against one another, give equality by strict convexity. Thus every subsequential limit is u, giving convergence of the full family weakly in H_0^1. Taking $w = u$ in the last inequality also gives a matching upper bound for the gradient norm, so convergence is in fact strong in H_0^1. □

4. Periodic forcing selects one periodic initial datum

Solution. For $f(t+\tau) = f(t), f \in L^\infty$, define the period map $P(g) = u(\tau; g)$ on $L^2(U)$. The difference of two solutions has zero forcing, and its energy estimate gives $\|P(g) - P(h)\|_2 \le e^{-\lambda_1 \tau}\|g - h\|_2$. Banach provides a unique g_*. The functions $u(t+\tau; g_*)$ and $u(t; g_*)$ solve the same problem with the same initial trace; linear uniqueness proves periodicity. Every periodic solution supplies a fixed point, proving uniqueness of the initial datum as well. □

5. A small gradient nonlinearity

Solution. Assume $f \in L^2$ and b is globally Lipschitz with constant L. Let Av solve $-\Delta Av = f - b(Dv)$, zero trace. Testing the difference of two outputs gives

$$\|D(Av - Aw)\|_2^2 \le LC_P\|D(v-w)\|_2\|D(Av - Aw)\|_2.$$

Thus $LC_P < 1$ makes A a contraction on H_0^1. Its fixed point is unique and belongs to H^2 by the linear L^2 regularity theorem. The value $b(0)$ affects the forcing bound but not the contraction factor. □

6. A positive solution above the first eigenvalue

Solution. Assume U connected, f bounded and Lipschitz, $f(0) = 0$, and $f'(0) > \lambda_1$. Normalize $\|\phi_1\|_\infty = 1$. For small $\varepsilon > 0$, differentiability at 0 gives $f(\varepsilon\phi_1) \ge \lambda_1\varepsilon\phi_1$, so $\underline{u} = \varepsilon\phi_1$ is a subsolution. Let $-\Delta\psi = 1$, $\psi = 0$ on the boundary, and take $\overline{u} = M\psi$ with $M \ge \sup|f|$. It is a supersolution and dominates $\underline{u}$ after increasing M: indeed $-\Delta[(\lambda_1\|\phi_1\|_\infty)\psi - \phi_1] \ge 0$, so comparison bounds ϕ_1 by a multiple of ψ. The ordered theorem yields $u \ge \varepsilon\phi_1 > 0$ in U. This pointwise statement uses the continuous representative, supplied by elliptic regularity with bounded right side. On a disconnected domain, $f'(0) > \lambda_1(U)$ alone need not give positivity on every component; connectedness is the needed qualification here. □

7. Classical comparison for the ordered iterates

Solution. With $h(s) = f(s) + \lambda s$ nondecreasing, each difference satisfies $(-\Delta + \lambda)(u_{k+1} - u_k) = h(u_k) - h(u_{k-1}) \geq 0$ once $u_k \geq u_{k-1}$. Its boundary data are zero, except at the first comparison where they are nonnegative by the subsolution trace. The maximum principle starts and propagates the induction. Likewise $(-\Delta + \lambda)(\overline{u} - u_{k+1}) \geq h(\overline{u}) - h(u_k) \geq 0$. Thus the entire chain lies between the barriers. The weak proof in §9.3 is its positive-part counterpart. □

8. Critical bubbles and lost compactness

Solution. For $n \geq 3$, $U(r) = (1 + r^2)^{-(n-2)/2}$ has

$$U' = -(n-2)r(1+r^2)^{-n/2},$$
$$-U'' - \frac{n-1}{r}U' = n(n-2)(1+r^2)^{-(n+2)/2}.$$

Thus $c = n(n-2)$. Scaling gives $U_\lambda(x) = \lambda^{-(n-2)/2}U(x/\lambda)$ and the same equation. For $2^* = 2n/(n-2)$, changing variables $x = \lambda y$ proves invariance of both $\|U_\lambda\|_{2^*}$ and $\|DU_\lambda\|_2$. These norms are finite: at infinity the respective integrands have radial orders r^{-n-1} and r^{1-n}, and they are regular at the origin. As $\lambda \downarrow 0$, $U_\lambda(x) \to 0$ for $x \neq 0$. Any strongly convergent L^{2^*} subsequence would have limit zero (by a further a.e. convergent subsequence), contrary to its constant nonzero norm. This proves noncompactness along a specified sequence; constant norms alone would not prove it. □

9. Energy and a weighted parabolic Pohozaev identity

Solution. Assume sufficient rapid decay for every displayed integral and boundary flux, and $F' = f$, $F(0) = 0$. Set $E = \int(|Du|^2/2 - F(u))$. Differentiate, integrate by parts, and use $u_t = \Delta u + f(u)$: $E' = -\int u_t^2$. For the second moment E_2 of the same density,

$$E_2' = -\int |x|^2 u_t^2 - 2\int u_t(x \cdot Du).$$

The two cross terms are $\int \Delta u(x \cdot Du) = (n-2)\int |Du|^2/2$ and $\int f(u)(x \cdot Du) = -n\int F(u)$. Therefore

$$E_2' = -\int(|x|^2 u_t^2 - 2nF(u) + (n-2)|Du|^2).$$

The signs differ because the two densities have different scaling degrees. Weighted integrability is essential; finite unweighted energy alone does not license differentiating E_2. □

集中需要极限证据

尺度不变范数只给有界性。还须取 $\lambda \downarrow 0$，证明几乎处处极限为零，再用强收敛保持范数得到矛盾。

10. The normal cone and metric projection

Solution. For a nonempty closed convex $K \subset \mathbb{R}^n$, its indicator energy has

$$\partial I(x) = N_K(x) = \{v : v \cdot (y - x) \le 0 \ (y \in K)\} \quad (x \in K),$$

and empty subdifferential outside K. Minimizing $I(x)+|x-z|^2/(2\lambda)$ gives $J_\lambda z = P_K z$, independent of λ, and $A_\lambda z = (z - P_K z)/\lambda$. The projection condition $z - P_K z \in N_K(P_K z)$ follows by varying along each segment to $y \in K$. At an interior point the normal cone is $\{0\}$; on a smooth boundary it is the outward normal ray. □

11. An irreversible flow

Solution. Take $H = \mathbb{R}$, $I(s) = |s|$. Then $\partial I(s) = \{\operatorname{sgn} s\}$ for $s \neq 0$ and $\partial I(0) = [-1, 1]$. Directly, $S(t)s = \operatorname{sgn}(s)(|s| - t)_+$ solves the inclusion a.e. For example $S(2)1 = S(2)2 = 0$ although $1 \neq 2$. This semigroup is contractive and continuous but its time maps need not be injective; a backward inverse flow cannot be defined on all states. □

12. A constrained sandpile as a normal-cone flow

Solution. Let $K = \{v \in L^2(\mathbb{R}^2) : |Dv| \le 1 \text{ a.e.}\}$, with the inequality understood in the weak-gradient sense. It is convex and closed in L^2: distributional limits preserve the gradient bound, by weak-* compactness of bounded gradients on compact subsets. Its indicator I is proper, convex and lower semicontinuous. Assume the proposed fields are regular and integrable enough that $f - u_t = -\operatorname{div}(aDu) \in L^2$ and integration by parts against $v - u$ is justified, for instance by compactly supported flux and approximation. For $v \in K$,

$$(f - u_t, v - u) = \int aDu \cdot (Dv - Du) \le 0.$$

Indeed where $a > 0$ one has $|Du| = 1$, so $Du \cdot Dv - |Du|^2 \le |Dv| - 1 \le 0$; elsewhere the integrand is zero. This is precisely $f - u_t \in N_K(u) = \partial I(u)$. The calculation is conditional on the asserted flux and regularity; it does not construct a from a solution of the abstract inclusion. □

13. Exponential decay of the speed

Solution. For a smooth solution with compatible boundary data, put $v = u_t$. Differentiating $u_t - \operatorname{div} DL(Du) = 0$ gives $v_t - \operatorname{div}(D^2L(Du)Dv) = 0$, with $v = 0$ on the boundary. Testing by v, uniform convexity and Poincaré give

$$\tfrac{1}{2}(\|v\|_2^2)' \le -\theta\|Dv\|_2^2 \le -\theta\lambda_1\|v\|_2^2.$$

Hence $\|u_t(t)\|_2^2 \le \|u_t(0)\|_2^2 e^{-2\theta\lambda_1 t}$ when the initial speed is finite. Otherwise the same estimate starts at any positive regular time, with a constant depending on that time. The uniform coefficient θ is the source of the decay rate. □

14. Total variation under nonlinear diffusion

Solution. Write $v = \phi(u)$ with $\phi' > 0$. Under the source's nonvanishing-gradient assumption and sufficient integrability and vanishing flux at infinity, $Du/|Du| = Dv/|Dv|$. Thus

$$\frac{d}{dt}\int |Du| = \int \frac{Dv}{|Dv|}\cdot D\Delta v.$$

Differentiating $|Dv|$ twice gives the pointwise Bochner identity

$$\Delta|Dv| = \frac{Dv}{|Dv|}\cdot D\Delta v + \frac{|D^2 v|^2}{|Dv|} - \frac{|D^2 v Dv|^2}{|Dv|^3}.$$

Its Laplacian integrates to zero under the stated flux hypothesis, so

$$\frac{d}{dt}\int |Du| = -\int \left(\frac{|D^2 v|^2}{|Dv|} - \frac{|D^2 v Dv|^2}{|Dv|^3}\right) \le 0.$$

The inequality uses $|D^2 v Dv| \le |D^2 v||Dv|$. This is the requested smooth calculation where denominators are defined. At critical points one needs a regularization and a limiting argument; no general BV contraction theorem is proved here, and nonvanishing and whole-space decay should not be silently imposed on arbitrary data. □

9.8 References

Source directions

Evans's §9.8 points to J.-L. Lions [L2] and Zeidler [Zd, Vol. 2] for monotonicity, and to [E1] for nonlinear weak-convergence methods. For fixed points the source recommends Zeidler [Zd, Vol. 1] and Gilbarg–Trudinger [G-T, Chapter 11]; for ordered methods, Smoller [S, Chapter 10]. The blow-up discussion follows Payne [Pa, §9]; the small-data theorem is attributed there to H. Fujita, J. Fac. Sci. U. Tokyo Sect. I 13 (1966), 109–124. The radial symmetry result is attributed to Gidas, Ni and Nirenberg, Comm. Math. Phys. 68 (1979), 209–243. The convex-flow material follows Brezis [Br2]; Barbu [Ba] and Zeidler [Zd, Vol. 2] are further directions for nonlinear semigroups. These are the source's attributions and bibliography keys, not a claim that all of those works have been independently consulted for this draft.

Proof dependencies and scope of this reconstruction

The finite-dimensional input is Brouwer from §8.1. Functional analysis inputs include weak compactness of bounded Hilbert-space sequences, Hahn–Banach separation, Sobolev and Rellich embeddings, and the Hilbert-space Picard theorem. PDE inputs are the linear elliptic and parabolic theories of Chapters 6–7, Sobolev truncation and time chain rules from Chapter 5, and convex integral lower semicontinuity from Chapter 8. The global boundary H^2 step in §9.1 explicitly uses the boundary difference-quotient argument of Chapters 6 and 8.

Editorial changes include a bounded-Jacobian hypothesis for that H^2 claim, an L^2 restart formulation for reaction–diffusion, a complete compact-convex-hull argument in Schaefer, extremal ordered solutions, a quantified small-support proof of moving-plane continuation, and a space–time resolvent test identifying the gradient-flow limit. The critical Fujita endpoint, general critical sign-changing Pohozaev nonexistence, arbitrary-data H^2 bounds down to time zero, general BV contraction, and reconstruction of a sandpile flux are not proved. English exposition, proofs, Chinese mathematical annotations, solutions, and TikZ diagrams are editorial reconstructions or additions. Compilation and explicit model checks do not constitute independent mathematical review.